Chapter 1, Problem 2P

Problem

Determine the current flowing through an element if the charge flow is given by

(a) $q(t)$ = (3) mC

(b) $q(t)$ = (4t2 + 20t − 4) C

(c) $q(t)$ = (15e−3t − 2e−18t) nC

(d) $q(t)$ = 5t2(3t3 + 4) pC

(e) $q(t)$ = 2e−3t sin (20π t) μC

Step-by-step solution

Step 1 of 5

(a)

The charge flow through an element is $q(t)=(3)\ \text{mC}$. The current flowing through the element is,

$$\begin{aligned} i &= \frac{dq}{dt} \\ &= \frac{d}{dt}(3) \\ &= 0\ \text{A} \end{aligned}$$

Therefore, the current flowing through the given element is $\boxed{0\ \text{A}}$.

Step 2 of 5

(b)

The charge flow through an element is $q(t)=(4t^2+20t-4)\ \text{C}$. The current flowing through the element is,

$$\begin{aligned} i &= \frac{dq}{dt} \\ &= \frac{d}{dt}(4t^2+20t-4) \\ &= (8t+20)\ \text{A} \end{aligned}$$

Therefore, the current flowing through the given element is $\boxed{(8t+20)\ \text{A}}$.

Step 3 of 5

(c)

The charge flow through an element is $q(t)=\left(15e^{-3t}-2e^{-18t}\right)\ n\mathrm{C}$. The current flowing through the element is,

$$\begin{aligned}
i &= \frac{dq}{dt} \\
&= \frac{d}{dt}\left(15e^{-3t}-2e^{-18t}\right) \\
&= \left(36e^{-18t}-45e^{-3t}\right)\ \mathrm{nA}
\end{aligned}$$

Therefore, the current flowing through the given element is $\boxed{\left(36e^{-18t}-45e^{-3t}\right)\ \mathrm{nA}}$.

Step 4 of 5

(d)

The charge flow through an element is $q(t)=5t^2\left(3t^3+4\right)\ \mathrm{pC}$. The current flowing through the element is,

$$\begin{aligned}
i &= \frac{dq}{dt} \\
&= \frac{d}{dt}5t^2\left(3t^3+4\right) \\
&= \frac{d}{dt}\left(15t^5+20t^2\right) \\
&= 75t^4+40t \\
&= 5t\left(15t^3+8\right)\ \mathrm{pA}
\end{aligned}$$

Therefore, the current flowing through the given element is $\boxed{5t\left(15t^3+8\right)\ \mathrm{pA}}$.

Step 5 of 5

(e)

The charge flow through an element is $q(t)=2e^{-3t}\sin(20\pi t)\ \mu\mathrm{C}$. The current flowing through the element is,

$$\begin{aligned}
i &= \frac{dq}{dt} \\
&= \frac{d}{dt}2e^{-3t}\sin(20\pi t) \\
&= 2\left[\frac{d}{dt}\left(e^{-3t}\right)\sin(20\pi t)+e^{-3t}\frac{d}{dt}\sin(20\pi t)\right] \\
&= 2\left[-3e^{-3t}\sin(20\pi t)+20\pi e^{-3t}\cos(20\pi t)\right] \\
&= e^{-3t}\left[40\pi\cos(20\pi t)-6\sin(20\pi t)\right]\ \mu\mathrm{A}
\end{aligned}$$

Therefore, the current flowing through the given element is

$\boxed{e^{-3t}\left[40\pi\cos(20\pi t)-6\sin(20\pi t)\right]\ \mu\mathrm{A}}$.